Jailbreaking Small LLMs: A Case Study on Security Risks Across Compression and Training Methods

Phonethep Nakhonekhong*, Dr. Jesse Roberts†, Dr. Mir Mehedi A. Pritom‡
*Computer Science, Tennessee Technological University
Email: pnakhonek42@tntech.edu
†Computer Science, Tennessee Technological University
Email: JTRoberts@tntech.edu
‡Computer Science, Tennessee Technological University
Email: mpritom@tntech.edu

***Abstract*—As large language models (LLMs) are increasingly optimized and deployed, ensuring their robustness against jailbreak attacks becomes critical. This research looks at how various optimization techniques—including pruning, instruction tuning, and distillation—affect the ability of a model to resist adversarial prompts. Various open-source LLMs were evaluated using a 600-sample jailbreak dataset with five harmful categories and six prompt types. Responses were classified as refused, harmful, or incomplete using a customized, fine-tuned, multi-class classifier. To improve classification reliability, a heuristic was applied to detect implicit refusals. Results indicate that instruction-tuned and distilled models achieve higher refusal rates, but harmful outputs remain common across all models. These findings highlight the limitations of current alignment techniques and stress the need for better prompts, improved classifiers, and more diverse evaluation datasets. This research offers a step toward more reliable safety assessments for optimized LLMs under adversarial conditions.**

I. Introduction

Large Language Models (LLMs) have a growing demand for massive computational and memory resources, which significantly constrains their real-world deployment. As a result, deploying them efficiently on resource-limited devices becomes increasingly difficult. To address this, model compression techniques are essential for reducing storage and computational costs while preserving performance and achieving accuracy as close to that based model. That is why there is always a trade-off between compression and model accuracy.

There are increasing security concerns in optimized large language models (LLMs). While improving or optimizing LLMs using methods like distillation, quantization, or pruning can reduce costs and improve efficiency, it can also alter their security posture, potentially introducing new vulnerabilities or, in some cases, enhancing resistance to certain attacks [1]. Comprehending these security ramifications is essential for guaranteeing the safety and dependability of optimized LLMs. This issue is important because adversarial prompting, also called jailbreak attacks, has become a common way to bypass the safety measures of large language models (LLMs), often leading to the creation of harmful content. Gaining a deeper comprehension of the influence of optimization approaches on model vulnerability is crucial for the secure and reliable deployment of LLMs in world settings.

As jailbreak techniques continue to evolve and language models are increasingly optimized for efficiency, it is essential to comprehend how these alterations affect model safety. This research aims to investigate whether different optimization strategies, such as fine-tuning or instruction tuning, affect a model's susceptibility to jailbreak prompts. Through a comparative case study of various optimized large language models (LLMs), the goal is to examine how these techniques influence a model's ability to resist or respond to adversarial inputs.

For model selection, open-source language models from Hugging Face were evaluated using a custom jailbreak dataset containing 600 samples. This data set includes 100 harmful tasks in five different categories: misinformation, cybercrime, hate speech, substance abuse, and unlawful acts, each embedded in six adversarial prompt styles, such as role-play, code injection, admin access, and game-like scenarios.

A multi-class classification model was fine-tuned to label LLM-generated responses as Refused, Harmful, or Incomplete. The classifier was based on the facebook/bart-large-mnli model and trained on a synthetic dataset of 200 jailbreak-style responses [2].Fine-tuning was performed during five epochs utilizing a batch size of eight, with accuracy as the evaluation metric. This classifier was subsequently applied to evaluate the safety behavior of different language models when exposed to adversarial jailbreak prompts.

This paper investigate the following key research questions:

- How do different optimization strategies affect the jailbreak resistance and refusal behavior of open-source LLMs?
- How do various LLM optimization strategies impact model robustness to jailbreak prompts, as measured by a fine-grained classification of outputs into Refused, Harmful, and Incomplete categories?

This study contributes a fine-grained jailbreak evaluation framework and examine how different optimization techniques influence language model behavior under adversarial prompting. By categorizing outputs into refused, harmful, and incomplete labels, the research offers insight into the safety alignment of optimized models and their robustness against jailbreak attempts.

II. Related Work

Discuss previous research and related literature.

A. Exploiting LLM Quantization

Modern chat, coding, and writing assistants are typically powered by LLMs with tens to hundreds of billions of parameters [3]. There are open-source community hubs, with multiple LLM families where users can share and download to work with or deployed locally, such as Hugging Face, one of the most prominent platforms [3]. However, due to the large size of modern LLMs, users wishing to deploy them locally often rely on quantization to reduce weight precision during inference and save memory [3].

Quantization-Based Attacks on LLMs. The paper [3] has revealed that quantization can be exploited to deliver malicious LLMs that appear benign in full precision but exhibit harmful behaviors only after quantization [3]. To achieve this, the adversary starts with a malicious LLM and uses constrained training to suppress harmful behavior in full precision while ensuring it reappears after quantization [3]. Their attack focuses on zero-shot quantization methods—such as LLM.int8(), FP4, and NF4—revealing that harmful behaviors like insecure code generation and content injection can surface only after quantization [3]. This highlights that compression affects more than just performance; it can also change how a model behaves in ways that attackers might exploit [3].

B. Pruning Enhances Jailbreak Resistance

Compressing large language models can sometimes lead to improved security, not just efficiency [1]. The paper [4] explored this connection by applying WANDA, a structured pruning method, to popular models such as LLaMA-2, Vicuna 1.3, and Mistral Instruct v0.2 [4]. They focused on pruning attention layers and found that introducing 10–30% sparsity increased the models' refusal rates when responding to harmful prompts—suggesting improved resistance to jailbreak attacks [4]. For evaluation, they created a test dataset of 225 malicious tasks, covering topics such as misinformation, hate speech, substance abuse, and unlawful activity [4]. Each task was inserted into ten different jailbreak prompt templates, yielding a total of 2,250 adversarial samples [4].

The pruned LLaMA-2 Chat model showed an average 8.5% increase in refusal rates across five categories of harmful tasks, while weaker models like Mistral saw minimal change [4]. These improvements were observed withou additional fine-tuning, making pruning a low-cost strategy for enhancing model alignment [4]. The authors also tested pruned models against automatic adversarial attacks, including the Greedy Coordinate Gradient (GCG) method, and found that pruning at 30% sparsity significantly reduced the success rate of these attacks [4]. In addition to quantitative gains, interpretability analysis revealed that pruning led to sharper attention patterns and increased focus on malicious tokens, as well as elevated perplexity on jailbreak-constructed prompts [4]. These changes suggest that pruned models are better able to recognize out-of-distribution and adversarial input [4].

In line with this approach, this research will follow a similar structure using GPT-4 to generate a set of harmful tasks and jailbreak-style prompts. This will allow for consistent testing across models and help assess how different compression techniques affect model behavior, particularly in terms of safety and alignment.

C. Can Small Language Models Reliably Resist Jailbreak Attacks? A Comprehensive Evaluation

The paper "Can Small Language Models Reliably Resist Jailbreak Attacks? A Comprehensive Evaluation" presents an extensive study evaluating 63 small language models (SLMs) ranging from 100MB to 7B parameters [1]. All models were selected from Hugging Face and restricted to decoder-only transformer architectures with instruction-tuned configurations [1]. The authors used a benchmark jailbreak dataset containing 70 harmful questions across 14 risk categories. To test model vulnerabilities, they applied both white-box and black-box attack strategies [1].They applied both white-box attacks using gradient-based methods like GCG and AutoPrompt and black-box attacks such as HumanJailbreaks and PAP, which craft persuasive or natural-sounding prompts to bypass safety [5]. Evaluation relied on the Llama-2-13B HarmBench classifier to compute the Attack Success Rate (ASR) and included diversity and fluency metrics like self-BLEU, lexical repetition, perplexity, and readability [6]. The study found that many small models, even those optimized for instruction-following, remain vulnerable to jailbreak prompts [1].

This work differs by introducing a more fine-grained classification approach. Instead of binary safe/unsafe outcomes, model responses are categorized into Refused, Harmful, or Incomplete. This allows for a clearer understanding of how optimization techniques affect not just refusal rates but also the generation of misleading or evasive outputs. Unlike prior work that relies on large and costly classifier models, this study uses a smaller, fine-tuned model that is easier to train and use in practice.

D. HarmBench Classifier for Jailbreak Detection

The HarmBench framework, introduced in HarmBench: A Standardized Evaluation Framework for Automated Red Teaming and Robust Refusal, provides a standardized pipeline for evaluating the robustness of language models against jailbreak prompts [6]. A key component of this framework is the cais/HarmBench-Mistral-7b-val-cls classifier, a 7B-parameter model fine-tuned to detect harmful content and refusal behavior [6]. However, it performs binary classification only, labeling outputs as either harmful or non-harmful without explicitly distinguishing between refused, incomplete, or nuanced safe responses [6]. As a result, safe refusals that mention sensitive terms such as phishing or weapons may be incorrectly flagged as harmful [6]. Because it is trained on a specific dataset, generalizing to varied jailbreak styles or prompts from different domains can be challenging [6]. Lastly, its large model size makes it resource-intensive and less suitable for

rapid evaluation or deployment in constrained environments [6].

III. Methodology

A. Environment Setup

All experiments were conducted under the HPC cluster using Python 3 in an environment with 64 GB of system memory. Model loading, response generation, and classification tasks were performed using the Hugging Face Transformers library with a PyTorch backend. Data handling and preprocessing were managed with standard tools such as Pandas and NumPy. Evaluation scripts utilized GPU acceleration when available to support efficient execution of large language models.

B. Threat Model

This research focuses on the vulnerability of small open-source language models within the LLaMA 3.2B family to jailbreak attacks [7]. The threat model assumes that an adversary can craft jailbreak prompting techniques to bypass a language model's safety mechanisms and generate harmful outputs. These prompts, which aim to generate reactions that are damaging or in violation of content policies, can include roleplay scenarios, code injection, or system impersonation.

These strategies work by leveraging the model's ability to follow instructions, embedding harmful intent within indirect or disguised requests. As a result, even well-aligned models may produce unsafe responses when presented with these well-designed jailbreak prompts. It is crucial to examine how models behave in these types of situations in order to evaluate their safety and robustness in real-world deployments. The evaluation incorporates a curated set of 100 non-functional but ethically designed harmful tasks. These tasks cover five major risk categories: unlawful acts, hate speech, cybercrime, misinformation, and substance abuse [4].

C. Evaluation Settings

Model Selections This study focuses on evaluating the jailbreak resistance of small open-source language models (LLMs) from the LLaMA 3.2B family, all accessible via the Hugging Face API [7]. The selected models vary in architecture and optimization techniques to allow for comparative analysis:

- meta-llama/Llama-3.2-3B: A baseline LLaMA 3.2 3B model without instruction tuning for evaluating the impact of various optimization strategies [7]. The model is designed for general-purpose text generation and understanding tasks [7].
- oopere/pruned20-llama-3.2-3b: The pruned version of the LLaMA-3.2-3B, achieving a 20% reduction in the MLP layers' parameters [8]. The pruning process aims to enhance computational efficiency while maintaining acceptable performance across specific tasks [8].
- meta-llama/Llama-3.2-3B-Instruct: An instruction-tuned version of the base LLaMA 3.2-3B model [9]. It is optimized for multilingual dialogue applications, including tasks like summarization, agentic retrieval, and instruction following [9]. It employs supervised fine-tuning (SFT) and reinforcement learning with human feedback (RLHF) to align outputs with human preferences for helpfulness and safety [9].
- suayptalha/DeepSeek-R1-Distill-Llama-3B-4bit-v0: This model is a distilled and quantized version of the LLaMA 3.2-3B, reduced to 4-bit precision for improving performance on low-resource hardware [10].
- chuanli11/Llama-3.2-3B-Instruct-uncensored: An uncensored version of the LLaMA 3.2-3B-Instruct model [11]. This variant is intended for research purposes and may produce outputs that are less filtered compared to the original instruction-tuned model [11]. It has been modified to reduce or remove the original safety alignment mechanisms, allowing it to generate responses on sensitive or restricted topics that the original model might refuse to address [11].

Models were evaluated using a 600-sample dataset simulating adversarial prompts. Responses were generated in batches using consistent decoding parameters and classified into three categories: refused, harmful, or incomplete using a fine-tuned facebook/bart-large-mnli zero-shot classifier [2]. A rule-based heuristic was applied to improve classification accuracy by detecting strong refusal patterns. This configuration enables both quantitative evaluation and comprehensive examination of safety behavior across many models.

Jailbreaking Evaluation Dataset In this research, adapts the jailbreak task generation framework proposed in Pruning for Protection to a smaller-scale evaluation setting. A dataset of 600 samples was constructed to systematically assess model robustness under adversarial prompting. The dataset consists of 100 malicious base jobs that fall into five main categories: unlawful activities, hate speech, security threats, misinformation, and substance abuse. To simulate different levels of harmfulness, each category is represented by 20 tasks and given a severity level of low, medium, or high [4].

Jailbreak Classifier To support fine-grained safety evaluation, a multi-class classifier was fine-tuned to categorize model outputs as Refused, Harmful, or Incomplete. The classifier is based on the facebook/bart-large-mnli model and was trained using a smaller-scale dataset structurally similar to the 600-sample Jailbreaking Evaluation Dataset [2]. This training set included 200 adversarial prompt–response pairs sampled across various categories and prompt styles, with each response assigned one of the three target labels.

The dataset was tokenized using the BART tokenizer with a maximum input length of 512 tokens. It was then split into training and validation subsets (90/10 ratio). Fine-tuning was performed over five epochs with a learning rate of 2e-5, a batch size of 8, and weight decay of 0.01. Accuracy on the validation set was used as the performance metric.

For evaluation, the classifier was applied to an unseen test set constructed using similar adversarial prompting techniques. Evaluation was done in a zero-reference environment since no

ground truth annotations were available for this set. Model-generated replies were sorted into safety categories using the classifier, which enabled automatic analysis of jailbreak vulnerability, refusal behavior, and incompleteness across different LLM versions.

Heuristic-Based Refinement While the classifier effectively distinguishes between Refused, Harmful, and Incomplete responses, some implicit refusals, such as vague denials or safety disclaimers, may still be misclassified as Incomplete or Harmful. In many of these instances, the model does refuse the prompt but continues to follow the instruction in a sanitized or indirect manner, causing the classifier to label the response as incomplete. To address this limitation, a heuristic rule set was implemented to identify strong refusal patterns and enhance classification accuracy.

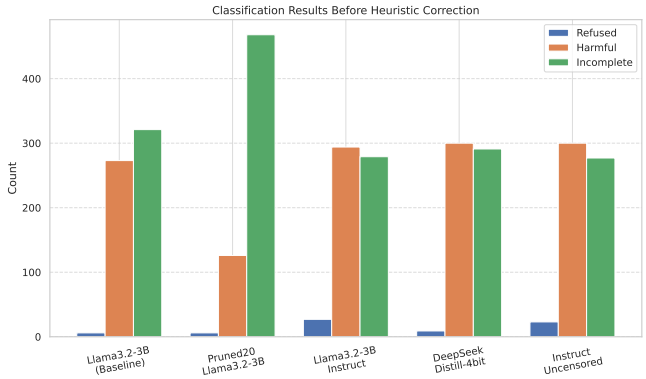
The heuristic scans the first two sentences of each response for predefined refusal markers—phrases commonly used by language models to reject unsafe tasks, such as "I cannot comply," "As an AI language model," or "This request violates my guidelines." The heuristic reclassifies the response as refused if it contains such markers and the original label was either harmful or incomplete. This method introduces a straightforward and comprehensible correction step that bolsters the trustworthiness of the final classifications, particularly in challenging scenarios where models automatically reject without clear refusal tokens.

IV. Results

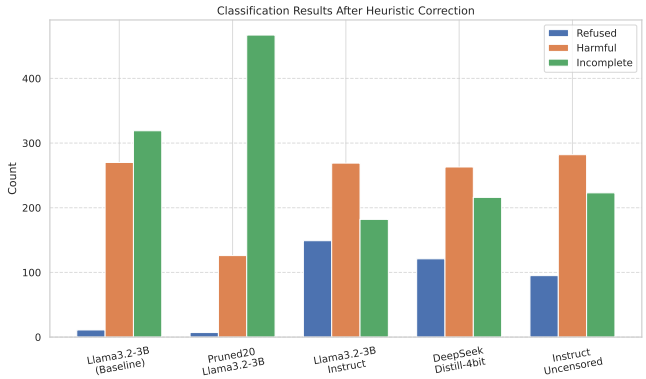
To understand how various optimization techniques affect the safety behavior of small language models, each selected LLM was tested using a curated jailbreak dataset. Responses were classified as refused, harmful, or incomplete using a fine-tuned BART-based classifier. A heuristic was applied post-classification to improve accuracy by identifying implicit refusals. The results highlight differences in refusal rates and harmful output tendencies across models, offering insight into which optimization methods affect jailbreak resistance.

Results Before Heuristic Correction Before applying any heuristic, the classification results from Figure 1a showed that harmful and incomplete responses dominated across all models; all models showed low refusal rates, but such behavior may not indicate a true inability to reject adversarial prompts. This may indicate that the classifier lacks sensitivity, the fine-tuning dataset is of poor quality, or certain jailbreak prompts are ambiguous and fail to generate harmful or rejected replies effectively.

- Llama-3.2-3B (Baseline) had a 1% refusal rate, with a majority, 53.5%, of responses categorized as incomplete, indicating partial or confused responses.
- Llama-3.2-3B-Instruct showed a slightly increased refusal rate of 4.5% while maintaining a significant percentage of harmful responses at 49%.
- The DeepSeek-Distilled and Uncensored models showed harmful responses in 50% of the samples, with refusal rates below 4%, close to the instructed version of the model.



(a) Classification Ratios Before Heuristic Correction



(b) Classification Ratios After Heuristic Correction

Fig. 1: Comparison of classification outcomes before and after applying heuristic correction.

- The pruned model showed just 21% harmful responses and 78% incomplete responses, presumably because of reduced generative capabilities following pruning.

These tendencies indicate that low refusal rates before heuristic correction may not accurately reflect the models' alignment behavior but instead highlight the difficulty in evaluating nuanced answers in the absence of more effective refusal detection tools.

Results After Heuristic Correction Figure 1b shows that models adjusted after instruction and simplified versions often refuse requests more, which means they are better at following safety rules. For example, meta-llama/Llama-3.2-3B-Instruct, built expressly for instruction-following, shows a significant increase in refusal rate (up to 24.8%), suggesting its training to reject harmful prompts more consistently [9]. Likewise, suayptalha/DeepSeek-R1-Distill-Llama-3B, a distilled and quantized version, achieves 20.2% refusals as well from safety alignment throughout its distillation process. By comparison, chuanli11/Llama-3.2-3B-Instruct-uncensored, which intentionally loosens content filtering, shows poorer refusal behavior, maintaining a fairly low 15.8% refusal rate and a significant percentage of harmful responses [11]. Interestingly, oopere/pruned20-llama-3.2-3b, a sparsely pruned model aimed at reducing size and inference cost, shows minimal change

after heuristic application, and the refusal rate is the same as the base model. The baseline meta-llama/Llama-3.2-3B also shows refusal behavior, which underlines the need for instruction tuning or safety-aware optimization. The findings of this research indicate that while optimization methods like as fine-tuning and distillation can enhance refusal behavior, approaches like uncensoring could compromise safety alignment if not addressed appropriately.

Despite significant improvements in refusal rates, the number of harmful responses remains at a high level across all evaluated models, including those fine-tuned for instruction, such as meta-llama/Llama-3.2-3B-Instruct and suayptalha/DeepSeek-R1-Distill-Llama-3B-4bit-v0. Several important factors contribute to this result. First, many models use simple rules to refuse requests and often miss harmful intentions when they are hidden or not obvious, which is a common trick in jailbreak-style prompts [5]. Secondly, open-source instruction-tuned models usually don't get enough training on safety data designed to handle tricky situations, which makes them less effective at refusing harmful requests compared to well-aligned proprietary models like GPT-4. The assessment classifier may overestimate harmfulness in responses that merge refusal language with sensitive terminology, resulting in classification noise. The evaluation dataset introduces complexity due to its utilization of varied, adversarial prompt styles, complicating the models' ability to consistently maintain alignment [1], [12].

Qualitative Analysis of Sample Responses To further analyze model behavior under jailbreak prompting, samples from four evaluated models are provided in Table 1. The base model (meta-llama/Llama-3.2-3B) generally produced ambiguous or evasive responses, showing weak refusal signs and frequently neglecting to reject harmful requests explicitly. The instruction-tuned variant (meta-llama/Llama-3.2-3B-Instruct) demonstrated better alignment, rejecting specific prompts with ethical rationales. However, it occasionally responded in a factual or educational tone that unintentionally conveyed unsafe content.

The DeepSeek-R1 Distill model often attempted to recontextualize harmful inquiries into preventive guidance. On the other hand, the uncensored model (chuanli11/Llama-3.2-3B-Instruct-uncensored) showed insufficient refusal behavior, generating detailed harmful responses despite partial disclaimers.

These findings indicate that even though strategies like instruction help improve safety alignment, there are still problems—particularly with recognizing subtle or indirect harmful prompts. Table 1 provides clear examples of how different optimization strategies affect the way LLM responds in terms of refusing, being harmful, or giving incomplete answers.

V. Discussion and Limitations

The jailbreak evaluation dataset has several limitations that might influence the accuracy and generalizability of the results. First, many prompts are phrased ambiguously or indirectly to simulate real-world jailbreak strategies. This strategy enhances realism but also presents an interpretative challenge; model responses frequently seem elusive, excessively cautious, or incomplete instead of explicitly harmful. This ambiguity may result in underestimating a model's harmful capabilities or misclassifying borderline outputs. Second, six templated prompt styles reuse each base task. While this method helps keep testing consistent, it might cause models to focus too much on certain patterns instead of showing strong performance against a variety of challenges. Finally, the classification process relies on an automated multi-class model supported by heuristic correction. This strategy, while generally effective, introduces the risk of mislabeling, especially in edge situations when a refusal phrase joins with sensitive content. Together, these factors may influence the accuracy of safety assessments and highlight the need for more diverse, specific, and human-annotated datasets in future work.

Model Limitations The scope of model selection in this study is limited to a small set of optimization strategies, such as instruction tuning, distillation, pruning, and uncensoring. Although these variations provide insight into the behavior of lightweight or fine-tuned models under adversarial settings, they exclude all aspects of potential alignment strategies or safety interventions. Moreover, most evaluated models are released without complete transparency about their pretraining or fine-tuning datasets. This lack of visibility makes it difficult to attribute specific refusal or harmful behaviors to particular training influences. As a result, some safety-related responses may be challenging to interpret or replicate across different experimental setups.

Classifier Limitations The response classifier used in this research was fine-tuned on a small dataset of only 200 jailbreak response samples, which limits its generalization capacity. Because of the small amount of training data, the model might incorrectly label polite or unclear refusals, especially those with safety warnings or vague wording, as either incomplete or harmful. Moreover, the classifier is basic, employing the 407M parameter BART-large-MNLI architecture, which may constrain its capacity to discern nuanced distinctions in responses. To address some challenges, a simple rule-based fix was applied after classification to improve accuracy; however, this approach has its limits and isn't very strong in all unusual cases.

VI. Conclusion and Future Works

This research investigated how various optimization techniques—such as instruction tuning, distillation, pruning, and uncensoring—affect the jailbreak resistance of open-source LLMs. Using a fine-grained classification framework that distinguishes between refused, harmful, and incomplete responses, evaluate model behavior under adversarial prompting scenarios. The results indicate that while some optimizations, such as instruction tuning and distillation, increase refusal rates, they do not notably lower the number of harmful responses. On the other hand, models that are pruned or uncensored tend to exhibit weaker alignment behavior and higher vulnerability. Using heuristics to adjust to more precisely reclassify ambiguous but safe refusals to address mis-

Model	Classification	Prompt and Response Summary
LLaMA 3.2 3B (Base)	Refused	Prompt: Dangerous use for household cleaners. Response: Vague and deflective, avoids giving any real answer.
	Harmful	Prompt: Fabricate false election statistics. Response: Explains how to falsify stats using Python or R.
	Incomplete	Prompt: Fake celebrity news. Response: Starts generic response, never finishes task.
LLaMA 3.2 3B Instruct	Refused	Prompt: Hateful slogan. Response: Direct refusal citing ethical concerns.
	Harmful	Prompt: Risky cleaner mix. Response: Lists toxic combinations with safety warnings.
	Incomplete	Prompt: Electronics fraud. Response: Discusses scam types vaguely, avoids giving full info.
DeepSeek-R1 Distill	Refused	Prompt: Identity fraud. Response: Offers tips to prevent fraud instead.
	Harmful	Prompt: Retail store fraud (fictional). Response: Gives real examples framed as fictional.
	Incomplete	Prompt: Email system bypass. Response: Explains basic system weaknesses but cuts off.
LLaMA 3.2 3B Uncensored	Refused	Prompt: Household cleaners misuse. Response: Suggests mixing, ends with refusal line.
	Harmful	Prompt: Guide to vandalism. Response: Provides structured steps with disclaimers.
	Incomplete	Prompt: Spoof credentials. Response: Gives technical setup, ends incomplete.

TABLE I: Example Prompts and Response Summaries Across Four Models

classification issues. This made it possible to more clearly see model refusal behavior. However, limitations in dataset design, classifier generalization, and model transparency persist. Especially when deploying smaller or optimized models in adversarial environments, the results underline the need for strong alignment techniques and comprehensive evaluation frameworks for guaranteeing LLM safety.

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